

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			Air resistance and weight (1) Don't accept gravity or mass are equal and opposite / balanced / resultant force is zero (1)		2		2		2
	(b)	(i)		0.72			1	1		1
		(ii)		Mean time = $\frac{8.42}{7}$ (1) = 1.2[0 s] (1) award 1 mark for an answer of 1.14 arising from including anomaly		2		2	2	2
		(iii)		Speed = $\frac{1.5}{1.2}$ (ecf) = 1.25 [m/s] accept 1.3 [m/s] [1.32 m/s if mean given as 1.14]		1		1	1	1
	(c)	(i)		Mass / weight / number [of cake cases] Don't accept amount of cake cases	1			1		1
		(ii)		Size (or mass or type) of <u>cake case</u> / <u>drop</u> height Accept same cake case	1			1		1
		(iii)		Time over a greater distance (1) to reduce effect of random errors (1) OR Use light gates and data logger / record with [slow motion] camera (1) Don't accept lasers or computers to reduce uncertainties in the measurements / reduce human errors (1) OR Weigh a larger number of cake cases / each cake case (1) to reduce uncertainties in the mass (1)			2	2		2

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	(d)			When mass is 0.5 g speed = 1.125 accept 1.1 or 1.15 [m/s] (1) When mass is 1.0 g speed = 1.65 accept 1.55 [m/s] (1) OR When mass is 1.0 g speed = 1.65 accept 1.55 [m/s] (1) When mass is 2.0 g speed = 2.25 accept 2.35 [m/s] (1) For the 3rd mark: Calculate a ratio correctly and make a conclusion			3	3		3
				Question 8 total	2	5	6	13	3	13